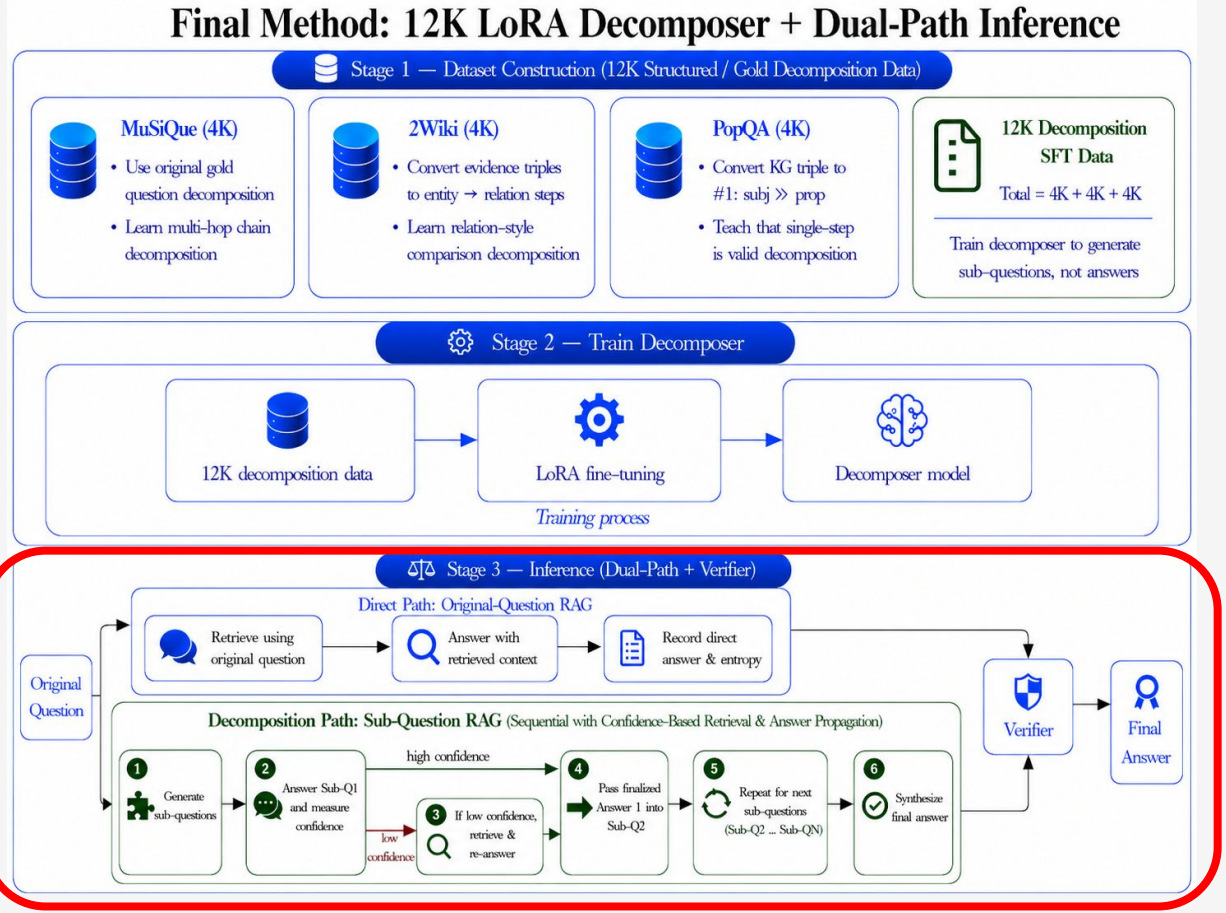
Progress Report

Hung-Yu Chien

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National Tsing Hua University
Intelligent Knowledge Management Lab.

Method - overview

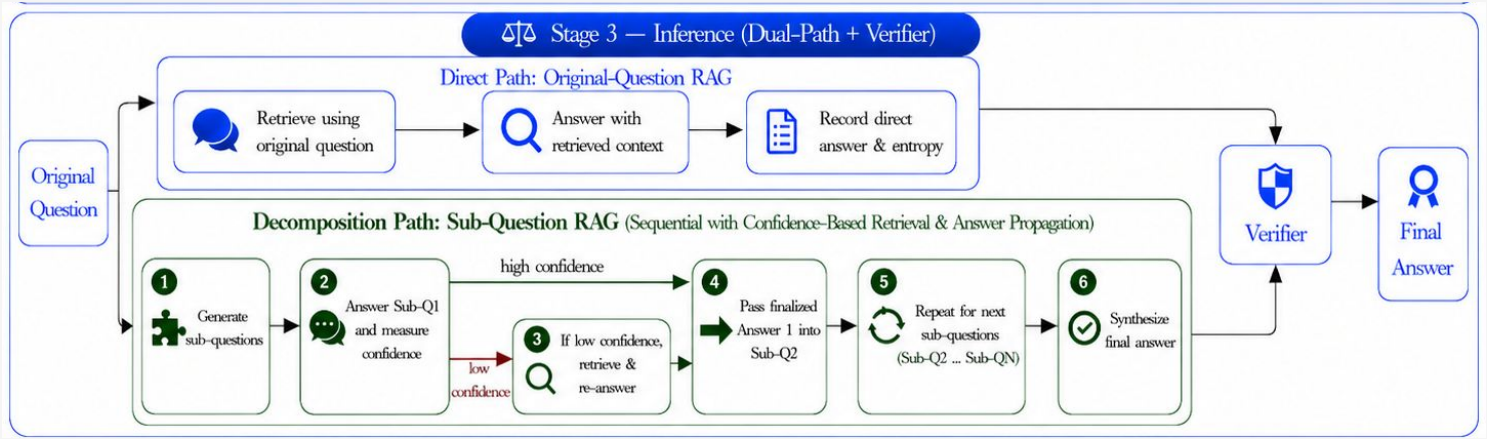


Method - overview

- Generate two candidate answers:

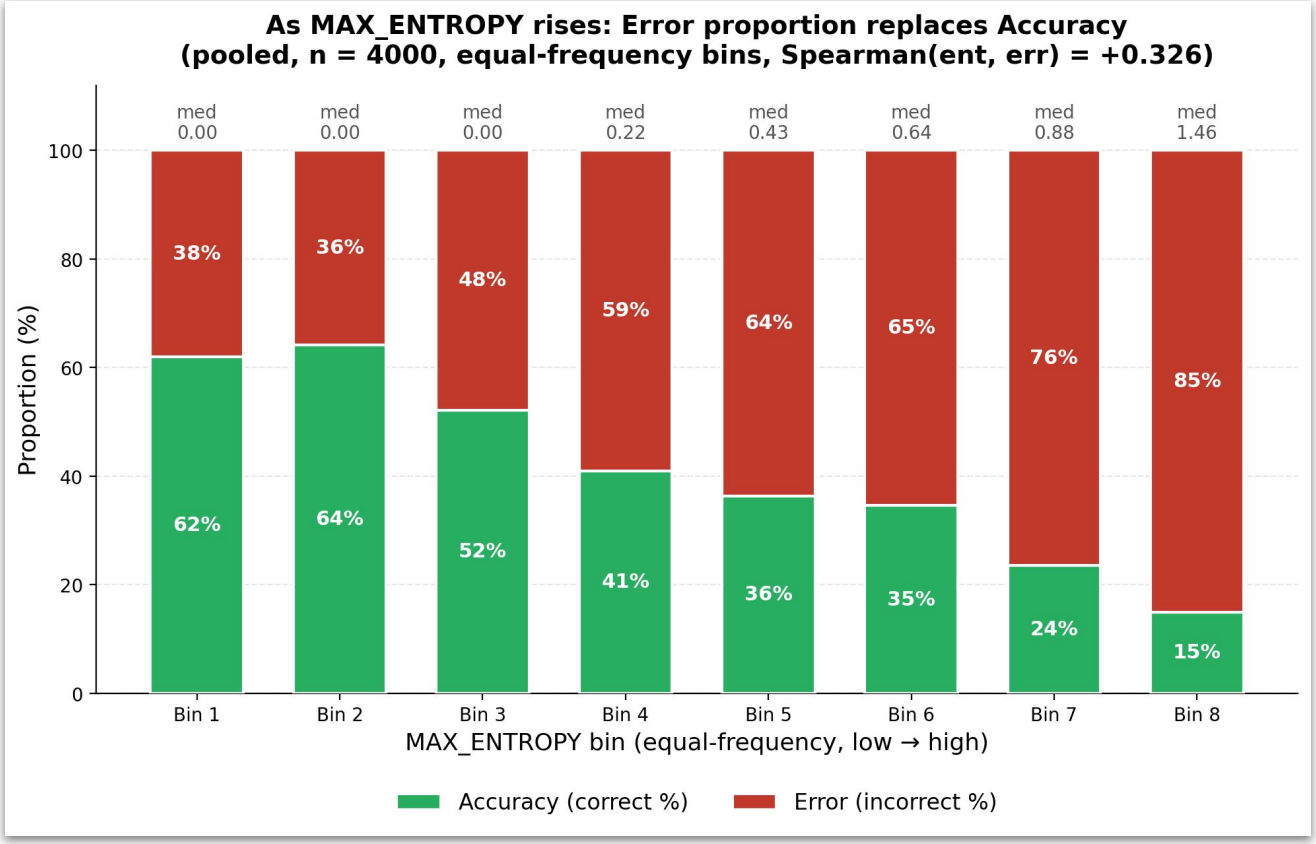
- Check three signals:
  - Direct-answer confidence
  - Decomposition stability
  - Direct vs. decomposed answer match

- Routing rule:
  - Choose direct if it is confident and decomposition is unstable
  - Otherwise choose decomposition



Recap - Correlation Between Entropy and Accuracy

Probing the Knowledge Boundary: Entropy as an indicator of model uncertainty.



Research - Dataset-level AUROC comparison

- After decomposition, the signal does not uniformly degrade
- 2Wiki sub-questions often show confident-but-wrong behavior.

Dataset	query-level (Original direct answer)	step-level (Sub-answer after decomposition)
HotpotQA	0.813	**0.696**
2Wiki	0.781	**0.540**
MuSiQue	0.716	**0.809**
PopQA	0.748	**0.749**

Example

2Wiki step-level is close to 0.5, meaning many wrong answers are confident-but-wrong.

Question	max_ent	parametric answer
A	3.2	✗ Wrong
B	2.5	✓ Correct
C	1.8	✗ Wrong
D	0.4	✓ Correct

AUROC = 3 / 4 = 0.75

(A, B): 3.2 > 2.5 ✓

(A, D): 3.2 > 0.4 ✓

(C, B): 1.8 > 2.5 ✗

(C, D): 1.8 > 0.4 ✓

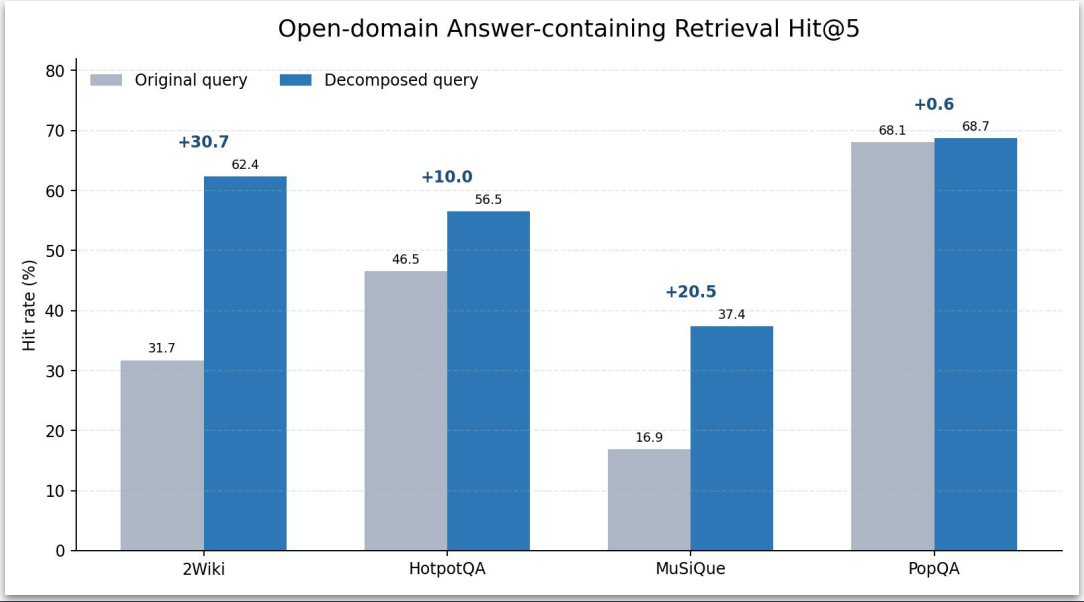
Research - Results of different approaches on three multi-hop datasets.

- SEEK achieves the best average result across the four datasets, improving over DeepRAG by +2.07 EM and +1.86 F1.
- The improvement mainly comes from 2Wiki, MuSiQue, and PopQA, with the largest gain on PopQA.

Llama-3-8B-Instruct										
Method	PopQA		Hotpot QA		2WikiMultihopQA		MuSiQue		Avg	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
CoT	21.20	25.33	27.20	37.75	28.20	34.85	13.70	22.97	22.58	30.23
FLARE	28.80	40.61	23.80	32.88	30.30	37.45	14.50	23.57	24.35	33.63
DeepRAG	42.50	47.80	**40.70**	**51.54**	48.10	53.25	22.50	30.40	38.45	45.75
SEEK (Ours)	**51.90**	**54.71**	35.50	46.11	**50.00**	**56.58**	**24.70**	**33.06**	**40.52**	**47.61**

Research - Retrieval Recall

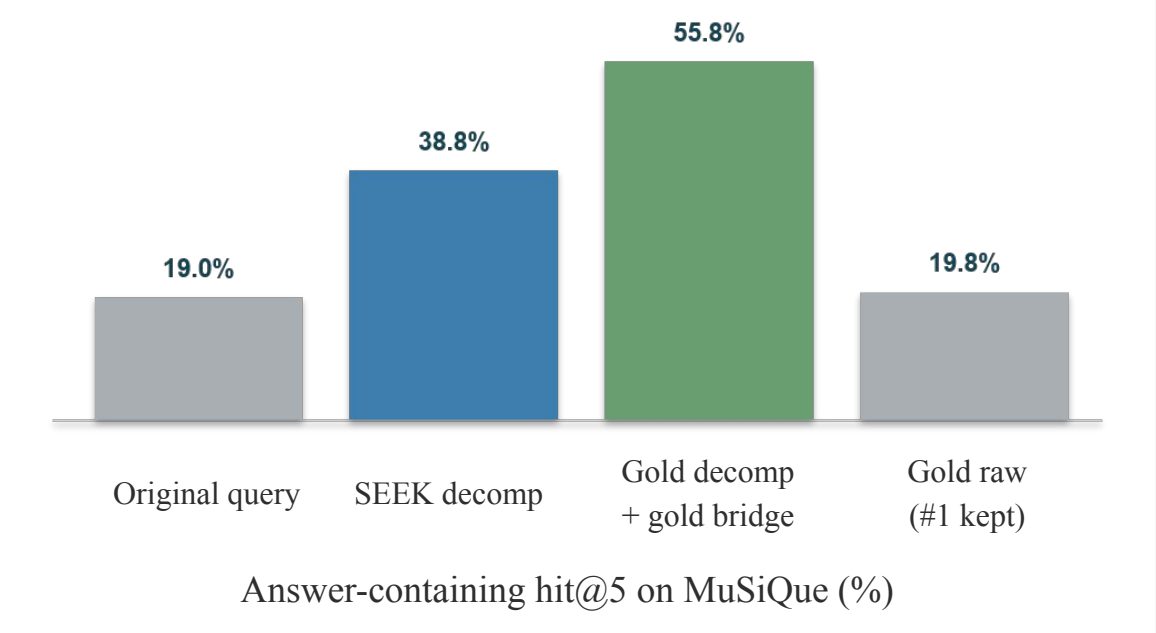
- **Goal**
  - Evaluate whether decomposition improves retrieval quality.
- **Metric**
  - Answer-containing hit@5 , Top-5 retrieved passages contain the gold answer.



Research - Oracle Decomposition Ceiling

Measuring the gap between SEEK decomposition and MuSiQue gold decomposition

- The benefit comes from using decomposition to guide step-by-step answering, not from simply generating decomposed questions.



Research - Where is the gap

Two isolation experiments separate the decomposition text from the intermediate answer. (2-hop example)

SEEK step-1 EM = 42.4% . When the bridge is correct , SEEK almost reaches the oracle.

Arm (hit@5)	Decomposition text	Intermediate answer	Result
SEEK	SEEK	Model Predicted	**38.8**
SEEK with answer	SEEK	Gold	**54.1**
Gold decomp	Gold	Gold	**55.8**

SEEK step-1 bridge	Count	SEEK hit@5	gold hit@5
Correct	424	**56.4%**	**58.5%**
Wrong	576	**26.3%**	**53.8%**

Research - Decomposer vs Base Model

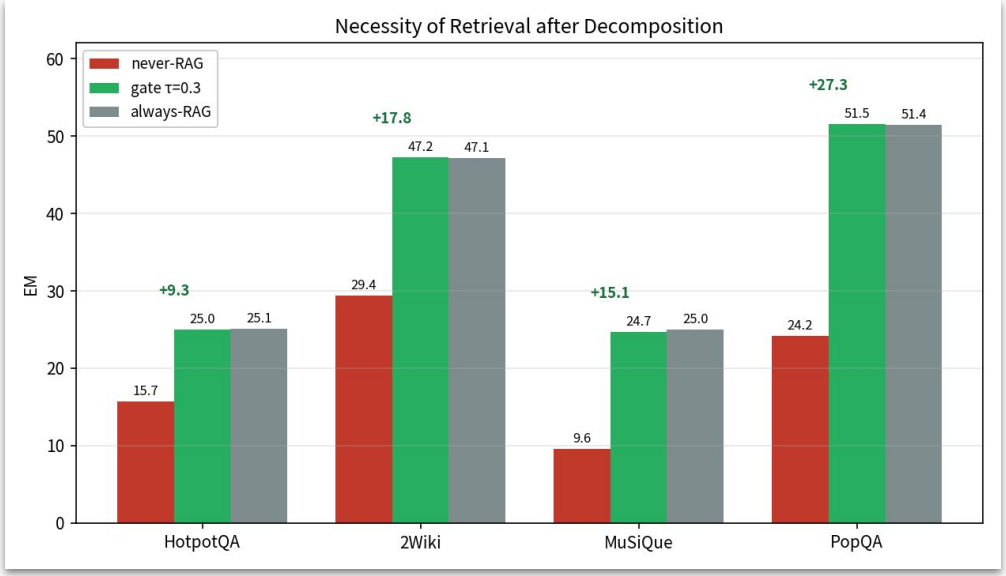
Using the same base model and prompt, we compare performance before and after training. (MuSiQue dataset)

	SEEK decomposer	**Base Model**
Correct decomposition	963 / 1000	149 / 1000
Format parsing failure	~0	153 / 1000
Query style	Short KG-triple queries	Long natural-language queries
Oracle-bridge hit@5	54.1	52.9 **(1.5× retrieval budget)**

Research - Retrieval Is Necessary After Decomposition

Measuring the benefit of retrieval: Parametric-only vs. Gate

- **without-RAG:** decompose only, no retrieval
- **gate:** retrieve only when entropy is high
- **always-RAG:** retrieve at every step



Research - Gating Improves Efficiency

We compare different retrieval thresholds τ in the final SEER configuration.
The goal is to check whether performance remains stable when the entropy threshold changes.

Dataset	Gate EM ($\tau = 0.3$)	Always-RAG EM	Always-Gate	Retrieval rate
PopQA	51.5	51.4	+0.1	99.3%
HotpotQA	25.0	25.1	-0.1	92.7%
2Wiki	47.2	47.1	+0.1	95.2%
MuSiQue	24.7	25.0	-0.3	90.8%

Research - Threshold Sensitivity

τ = 0.3 is selected as the final setting because it gives the best average EM/F1.
Higher τ values reduce retrieval cost with small accuracy drops.

Dataset	***τ*=0.3**		***τ*=0.5**		***τ*=0.8**		***τ*=1.0**	
	EM	F1	EM	F1	EM	F1	EM	F1
PopQA	**51.90**	**54.71**	51.90	54.71	51.70	54.38	51.80	54.41
HotpotQA	**35.50**	46.11	35.40	46.10	35.50	**46.17**	35.50	46.12
2Wiki	**50.00**	56.58	50.00	**56.65**	49.00	55.78	48.10	55.09
MuSiQue	**24.70**	**33.06**	24.50	32.61	23.70	31.84	22.60	30.91
Average	**40.52**	**47.61**	40.45	47.52	39.98	47.04	39.50	46.63
Avg retrieval rate	94.16%		91.32%		85.97%		82.35%	

Research - Can Direct Gate Replace Decomposition

Without decomposition, the gate is limited to choosing between direct answering and single-shot retrieval on the original question. In multi-hop QA, both options often fail.

Dataset	Dir-never	Dir-gate best-τ	Dec-gate (SEEK)	Δ
2Wiki	12.6	18.6	**47.2**	**+28.6**
MuSiQue	5.9	10.8	**24.7**	**+13.9**
PopQA	22.4	39.2	**51.5**	**+12.3**
HotpotQA	18.2	30.7	**25.0**	**-5.7**

Research - Path-aware Title Retrieval

- **Observation**
  - Short sub-questions usually point to a specific entity, so searching the title field helps find the right article.
  - Long original questions contain more words and constraints, so title matching can bring in irrelevant articles.
- **Strategy**
  - Direct path / Original question → Title off
  - Decomposition path / Sub-question → Title on

Dataset	Title-off		Title-on		Path-aware	
	EM	F1	EM	F1	EM	F1
PopQA	51.2	53.5	51.5	53.9	**51.8**	**54.2**
HotpotQA	31.9	41.2	30.5	39.9	31.8	**41.2**
MuSiQue	22.5	30.5	24.5	33.1	**24.6**	**33.5**
2Wiki	45.2	51.3	45.9	52.5	**46.9**	**53.5**

Action Item

Type	Action Item	Description	No.	Start	Comp.	Deadline	Status
Research	Experiment	Survey papers on question decomposition methods	E-23	6/19	---	6/26	In Progress
	Experiment	Evaluate our method across different models	E-24	6/20	---	6/26	In Progress
	Experiment	Analyze confidence scores in correct and wrong cases.	E-25	6/26	6/27	6/28	Complete
	Experiment	Analyze decomposition errors and confidence-based retrieval limitations.	E-26	6/28	7/1	7/1	Complete
	Experiment	Use a confidence-only baseline to show the benefit of decomposition.	E-27	7/1	7/3	7/3	Complete

Thanks!